Balaji Balachandran

balajib@stanford.edu
(+1) 630-943-2073
23417 W Lansdowne Road, Plainfield, IL
linkedin.com/in/balaji-balachandran-dev/
github.com/balaji-balachandran

Education

Stanford University GPA: 3.89/4.0

Master of Science in Computer Science w/ specialization in Network and Computer Security Sept. 2025 – June 2027

Relevant Coursework: Advanced Networking and Distributed Systems, Cryptography, Cryptocurrencies and Blockchains, Quantum Cryptography, Randomized Algorithms

University of Illinois Urbana Champaign GPA: 4.0/4.0

Bachelors of Science in Computer Science Aug. 2022 – May 2025

Relevant Coursework: Computer Security, Cryptography, Distributed Systems, Applied Parallel, Programming, Artificial Intelligence, Applied Machine Learning, Algorithms, Operating Systems, Networking, Linear Algebra, Computer Architecture, Internet of Things

Institutional Honors: University Honors, Bronze Tablet Scholar (highest UG honor), Dean's List 2022-2025

Experience

Empirical Security Research Group Oct 2025 – Present

Research Assistant, PI: Zakir Durumeric *Stanford, CA*

- Designing systems to improve efficiency of large-scale web measurement by reducing network transfers in automated crawlers.
- Built tooling to omit non-critical web resources, reducing bandwidth by 50% for 60% of sites while preserving fidelity.
- Formulated a novel modality-aware framework to classify web resources and evaluate network efficiency vs. site fidelity
- Implemented shared-socket architecture in Zgrab for multiplexed scanning from a single IP/port, improving connection reuse.
- Coauthored *"Stop Trying To Make Fetch Happen": Towards a Modality-Aware Understanding of Unnecessary Resources on the Web* to appear in IMC '26

LinkedIn June 2026 – Present

System and Infrastructure Engineer Intern *Mountain View, CA*

- Architected and independently built a dedicated microservice by decoupling traffic validation from disaster recovery, enabling continuous service deployment and isolated iteration.
- Implemented a Kalman filter algorithm and statistical analysis models to detect pre-failover anomalies and guarantee data robustness to mitigate failure during traffic shifts.
- Engineered a declarative simulation-testing framework to validate algorithm modifications offline, bypassing production risk and significantly accelerating feature development.

SysNet Lab May 2024 – Sept 2024

Research Assistant, PI: Radhika Mittal *Champaign IL*

- Used eBPF to trace requests at kernel-level across distributed microservices to enable observability without modifying code.
- Developed probabilistic techniques to associate microservice calls with requests across various concurrent threading models.
- Built a passive tracing model that reconstructs end-to-end request paths across services using observed network and execution patterns.

Course Assistant Jan 2024 – May 2025

Prob & Stats in CS [CS 361] and Algorithms [CS 374] – University of Illinois Urbana-Champaign *Champaign, IL*

- Conduct weekly office hours for two classes with 350+ students per semester, supporting students' understanding
- Supported two of the largest courses within the CS department required by every undergrad, helping students understand general statistics concepts as they pertain to computer science and understand how to methodically design algorithms

Projects

Communipute | *Python, React, Typescript, Flask, Docker, Kubernetes, JWT* February 2023

- Developed a crowd-sourced cloud service allowing users to donate and request additional compute power from their community. Implemented secure authentication through JWT, networking through polling, and a React+TypeScript frontend
- Winner of three track prizes at Stanford's TreeHacks 2023 hackathon (the world's largest collegiate hackathon with **1600+** participants)

Technical Skills & Honors

Languages: C/C++, Go, Python, Solidity, Bash, Swift, Kotlin, JavaScript, Typescript, HTML/CSS, MySQL
Frameworks: React, Node.js, Flask, React Native, Microsoft Azure, FastAPI, Selenium, CUDA, AWS, Koin
Honors: Winner of Stanford's Treehacks (2023), UIUC's HackIllinois (2023), and Synchrony's hackathon (2023)